

Final Project: Proposal/Dataset Selection

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The group members for our final project are Francisco Hernandez, Sri Ram Prabu Elenchezian, and Andre Shirevani Harandi. We, as a group, have decided to focus on Pitcher's Stuff and Injury Risk Prediction via an R built-in baseball library (Lahman), specifically working with the Pitching dataset table and further complementing it with Statcast metrics via a CSV file or baseballr library in R. We are aiming to explore the relationship between advanced pitching metrics and injury outcomes via the Lahman database which provides season-level pitching records for thousands of players dating back to 1871. While Statcast, available since 2015, provides millions of pitch-level observations. For feasibility, we plan to subset the data to recent MLB seasons (e.g., 2015-2023), which will allow us the opportunity to accurately combine both sets of information and work with a manageable dataset.

This project will not only allow us to generate strategic and operational insights (e.g roster management and injury prevention) while applying advanced R methods covered in this course but it will also permit us to understand if strikeout rate (SO9) and walk rate (BB9), velocity, pitch types and spin rate predict whether a pitcher will appear in the injured list the following year? What detrimental factors are affecting pitchers in today's game, and how can they be mitigated to improve their stamina in the game and overall longevity? Indeed, it is time to discover how pitchers can reduce injuries and become inning eaters once again while still maintaining the implementation of that "Flame Throwing Game".

Some possible analytical methods we might apply include logistic regression to predict IL status (a binary dependent variable) and predictors (such as walk rate, pitch velocity, strikeout rate, etc.). EDA, which will include descriptive statistics, correlation matrices, and other rich visualizations to explore different relationships among key variables. Multivariable regressions, where we will be modeling continuous pitching outcomes (e.g., ERA, innings pitched).

References

Lahman package | *RDocumentation*. (2023). Rdocumentation.org.

<https://www.rdocumentation.org/packages/Lahman/versions/12.0-0>

4 Statcast | *Fundamentals of Collecting and Analyzing Baseball Data*. (2015).

Bookdown.org. <https://bookdown.org/jjovanovic/jjovanovic/statcast.html>